Assignment 1

Group 24

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1 Part 1: Vertical variation of wind speed

1.1 Met Mast and Location

Task 1: Describe the met mast and its location. How do the surroundings of the mast look in terms of terrain roughness? How far is the mast from the turbine and in which direction? Why?

The measurements are from DTU Risø Campus using a met mast with a total height of 70 m. The mast is around 120 m, away from the Vestas V52 wind turbine. Regarding the terrain, the location is characterized by varying roughness lengths. To the west, it's dominated by the Roskilde Fjord, which leads to very low terrain roughness. In the other directions, the roughness is higher due to a mix of open fields with light vegetation and the campus buildings. The mast is positioned at this distance and orientation because the prevailing winds in the area come from the west, over the fjord. This setup allows for the measurement of undisturbed wind conditions, as the mast is located upwind of the turbine, ensuring that the data collected is not influenced by the wake effects of the turbine. Therefore free flow measurements can be obtained.

1.2 Data Collection

Task 2: Describe the data collected. What has been measured, what units, what sample rate, where at the mast and with what instruments?

The data collected on February 10th, 2026 between 10.00 and 11.00, corresponds to three categories: **Wsp, AirAbs, and Sdir**. They respectively describe horizontal wind speeds measured in m/s^2, temperature measured in $^\circ$ C, and southern wind direction measured in $^\circ$.

Since wind is a dynamic fluid, and it varies according to its distance from the ground and its temperature, each category varies with height. This encourages recordings for each category to happen simultaneously at different rungs of the mast. Wsp is measured at 70 m (i.e. the uppermost level of the mast), 57 m, 44 m, 31 m, and 18 m; then, AirAbs is measured at 70 m and 18 m; and lastly, Sdir is measured at 70 m, 44 m, and 18 m.

All three categories at their respective mast heights get 3000 readings every 10 minutes, i.e. the have a sampling rate of 5 Hz.

Lastly, the recordings at different heights also indicate the tool used to make the measurements since we know where in the mast of turbine V52 the tools are. As such, for wind speed (Wsp), the cup anemometer and the sonic anemometer are used, for temperature (AirAbs), the thermometer is used, and for wind direction (Sdir) only the sonic anemometer is used since the cup anemometer, which is also installed and the pertinent mast heights, cannot track wind direction.

1.3 Raw Time Series Plot

Task 3: Plot the raw time series of the wind speed at the different heights. What time scales do you see? Do they differ for the different heights?

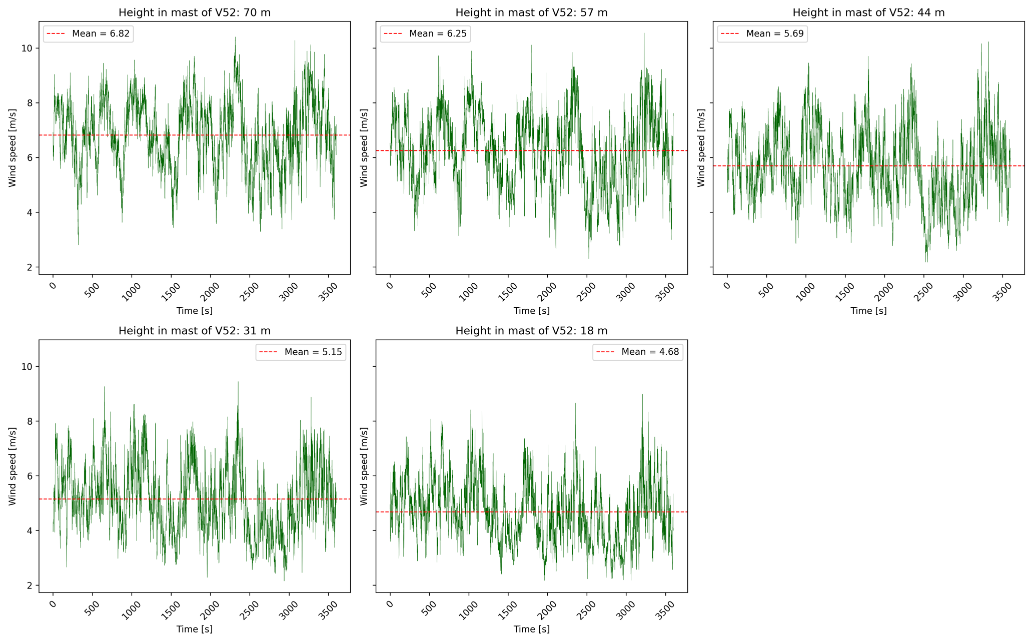

Figure 1: Raw wind speed time series at various heights (10th Feb 2026).

The plots in Figure cite the relevant figure show the wind speeds recorded at different heights, i.e. rungs, of the measurement mast of turbine V52. Across all five of them, one can observe irregularities in the behavior, which correspond to turbulences in the flow of the wind. These are described in two scalestime-wise and spatial, which in the plots correspond to the duration of the inflections in the curve and to the shape of the curve, respectively.
With this in mind, turbulences seem to occur in a shorter and in a longer time span at all five heights, while keeping a similar overall shape. The first relates to time scales, so changes in wind speed happen in the seconds-long scale, which correspond to local and small disturbances that look almost like noise, and so do they happen in the minutes-long scale, which correspond to overall pattern changes that lead to evident dips and raises in the plots.
The former scale then lead to the spatial scale, which determines what type of duration the turbulence will have. Sudden changes (seconds-long), like a sudden draft, will usually be physically contained an occur at a specific height, which allows us to have five different plots. However, progressive changes (minutes-long), like the transition from day to night, will usually span several heights and affect a range of air parcels, which results in still similar 5 plots. This is particularly evident in the 2400-3300 seconds range in Figure cite the relevant figure, which shows a w-shape in the five heights where an important drop is followed by a slight increase, a drop and an important increase in wind speed.
Lastly, height and average wind speed appear to be directly proportional, such that lower heights experience lower mean wind speeds while the opposite is true for higher heights.

1.4 Logarithmic Wind Profile

Task 4: Plot the average wind speed for each height against height z. Can the profile be described by a logarithmic wind profile?

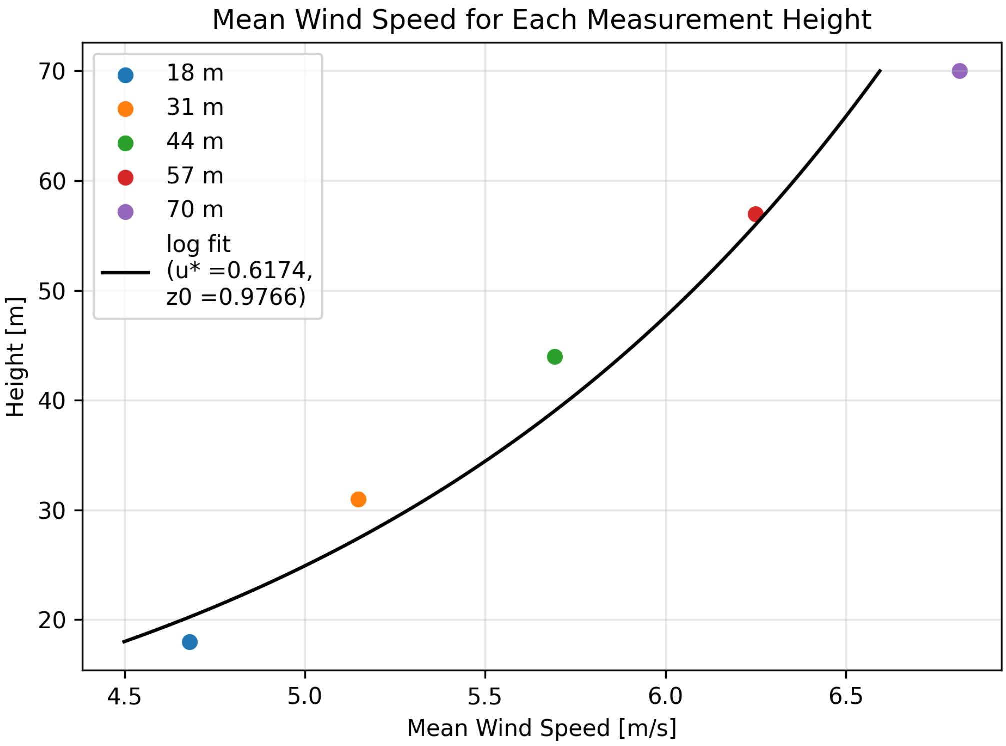

Figure 2: Average wind speeds at various heights and fitted logarithmic profile (10th Feb 2026).

In Q1 we observed that "the prevailing winds in the area come from the west, over the fjord...which leads to very low terrain roughness", yet we must recall that there existed a stretch of open land, recently snowed on, between the fjord and the measurement mast. According to the literature (page 14, slides of "Wind Profiles" Lecture), the roughness length of farmland with open appearance, snow surfaces, and fjords are 0.05, 10^{-3}, and 10^{-4}, respectively.
The fitted curve however, used $z_0 = 0.9766$, which more closely resembles the roughness of a city. This may be the case because during the day of the visit we had eastern winds, which corresponds to an influx from nearby populated areas like Ågerup or further and denser areas like Copenhagen.

1.5 Roughness Length and Friction Velocity

Task 5: Estimate the roughness length z_0 and the friction velocity u_* from the observed wind profile.

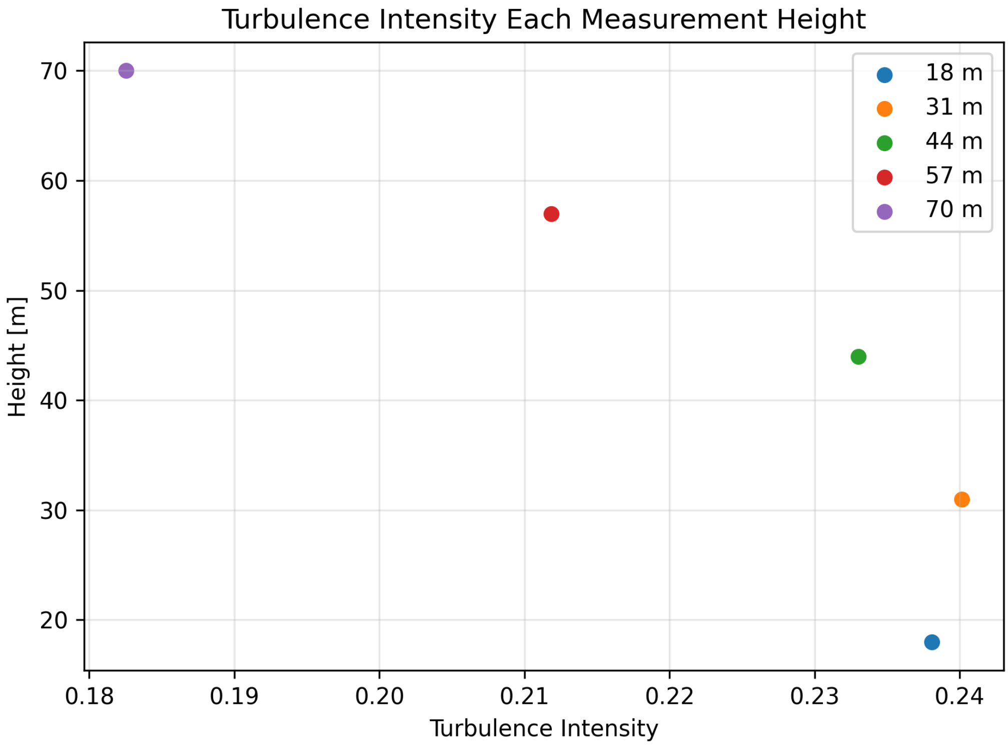

Figure 3: Turbulence intensity at various heights (10th Feb 2026)

Turbulence intensity is the ratio of the standard deviation and mean value of the wind speed. It is thus dimensionless, and it helps to quantify how "erratic" turbulences are.
Figure *figurenumber* shows a general indirect proportionality between height and turbulence: the greater the height at which the measurement was taken, the lower the turbulence intensity. There is, however, a slight deviation from this pattern at turbulence intensity measured at 31 m since it is higher than the turbulence intensity measured at 18 m.
The overall pattern is likely dictated by the disminution of obstacles and the homogeneity of temperature at greater heights. Immediately around turbine V52 there is no large objects, e.g. skyscrappers, dense forests, etc., so there is fewer occassions for disrupting wind flow. Then, the ground, which absorbs and later emits the most heat in the course of a day and supports objects (machinery, buildings, etc.) with their individual heat absorption and radiation, introduces a heat gradient that alters more the temperature of the wind at lower heights.
The former will be generally true and the single observable deviation in the decrease of turbulence intensity with speed at 31 m may come from an overlap with other large objects, greater wind drafts at that height, or ...

1.6 Stability Impact

Task 6: Estimate the lapse rate by using the data from the two temperature sensors at 18m and 70m. Compare its value with the dry adiabatic lapse rate and give a comment. Discuss the validity of the logarithmic profile under the observed stability conditions.
To estimate the atmospheric lapse rate during the measurement period, we use the two temperature sensors mounted on the mast at heights of 18 m and 70 m. First the mean temperatures

at both hights are computed. The resulting values are

$$T_{18} = -0.24\,^\circ\mathrm{C}, \qquad T_{70} = -0.61\,^\circ\mathrm{C}.$$

The lapse rate (vertical temperature gradient) is then obtained from

$$\Gamma_{\text{measured}} = \frac{\Delta T}{\Delta z} = \frac{T_{70} - T_{18}}{70\,\mathrm{m} - 18\,\mathrm{m}} = \frac{-0.61 - (-0.24)}{52\,\mathrm{m}} = -0.00722\,\mathrm{K/m},$$

which corresponds to

$$\Gamma_{\text{measured}} = -7.22\,\mathrm{K/km}.$$

For comparison, the dry adiabatic lapse rate is

$$\Gamma_{\text{dry}} = -9.8\,\mathrm{K/km}.$$

Our measured lapse rate is less negative (the temperature decreases more slowly with height) than the dry adiabatic value of $-9.8\,\mathrm{K/km}$. This indicates weakly stable stratification where vertical motions and turbulent mixing are suppressed compared to neutral conditions.
The logarithmic wind profile is valid only for neutral conditions, where buoyancy effects are negligible and turbulence is mainly driven by shear. In contrast, our measured lapse rate of $-7.22\,\mathrm{K/km}$ indicates weakly stable conditions where buoyancy suppresses vertical mixing. This can cause the wind profile to deviate from the ideal logarithmic shape. This means that the logarithmic fit used in Question 4 still works reasonably well over the mast height range, but the fitted roughness length and friction velocity may be slightly biased because stability effects are not directly included.

1.7 Atmospheric Stability Analysis and Reference Comparison

Task: With basis in the reference data, discuss if the conditions at the day of your measurements were stable, neutral or unstable. Use also the lapse rates calculated from the reference data as guidance.
To classify the atmospheric conditions during the visit on February 10, 2026, the measured data was compared against three standard reference datasets, representing stable, neutral, and unstable atmospheric stratifications.
The wind speed profiles are normalized by the velocity at 44 m (v/v_{44}) to allow for a shape-based comparison of the vertical shear regardless of the absolute wind magnitude.

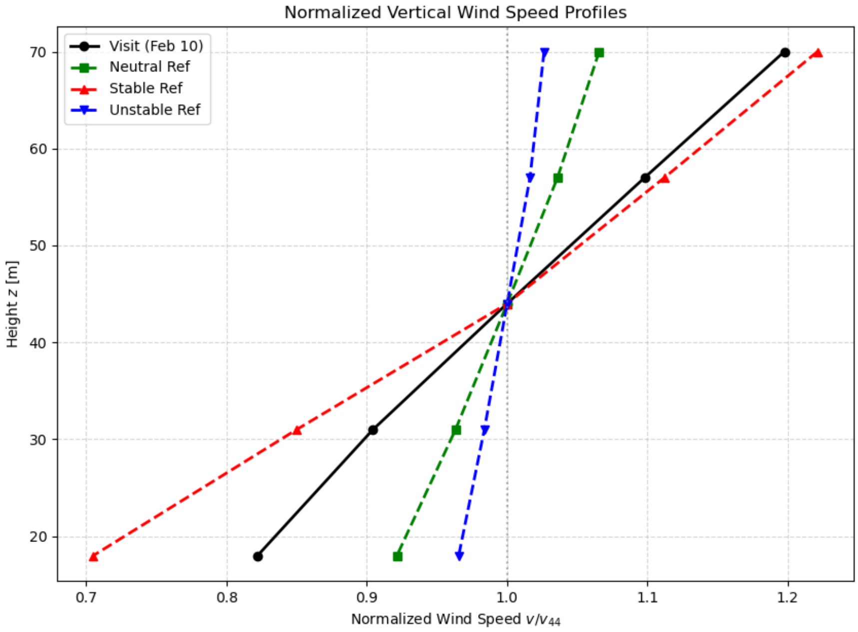

Figure 4: Normalized wind speed profiles v/v_{44} for the visit day compared to DTU reference datasets.

As seen in Figure **??**, the visit data exhibits a significant velocity gradient, which is similar to the stable one. In comparison, the unstable profile is much flatter due to better vertical mixing. Apart from the velocity profile, one can also look at the turbulence intensity (TI) profile to confirm the stability classification. Unlike the velocity, the TI values are plotted without normalization to compare the raw turbulence levels across different heights.

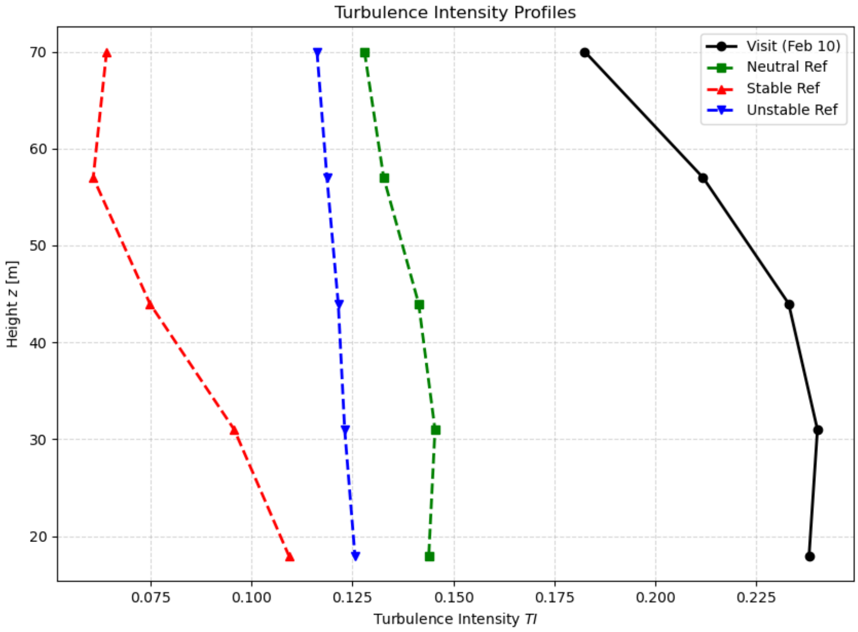

Figure 5: Turbulence Intensity (TI) profiles for the visit day compared to the reference datasets.

The turbulence intensity ($TI = \sigma_u/\overline{u}$) was calculated for all heights to characterize the mixing processes during the visit. The results show an overall average TI of 0.2211, with values ranging from 0.1825 at 70 m to 0.2402 at 31 m. Those values are quite high compared to the reference data. The shape resembles the stable reference and the neutral one most closely, having higher turbulence near the ground and decreasing with height.
To compare the data from the visit with the reference dataset, one can look once again at the lapse rates. In Table **??**, the calculated lapse rates for the visit and the reference datasets are summarized. The visit's lapse rate of -7.22 K/km is less negative/more stable than the neutral reference and less stable than the stable refernce, indicating a fairly stable condition between those two references, but closer to the stable one. That fits well to the visual comparison of the velocity and turbulence profiles, which also suggest a stable stratification during the visit.

Table 1: Calculated Lapse Rates and Stability Classification.

Dataset	**Lapse Rate [K/km]**
Visit (Feb 10)	-7.22
Neutral Ref	-11.10
Stable Ref	2.22
Unstable Ref	-13.54

Since the measured lapse rate is also greater (less negative) than the dry adiabatic rate of -9.8 K/km, one can conclude that the atmosphereric conditions during the visit were stable.

1.8 Weather Conditions

Task 8: How was the weather at the day of visit? Does the observed weather support your finding of question 7?

The day we visited Risø was windy from the eastern direction with some small snowflakes. Generally, the site experiences western winds from the fjord, but that was not the case that day. The ground was covered with snow and the temperature was a bit under zero degrees Celsius while the sky was full of bright grey clouds. Since the wind came from the land side, it seemed like very cold wind. These observations support the outcome from Question 1.7, confirming a stable atmospheric stratification. The presence of snow cover on the ground is a driver for stability, as it increases surface albedo and causes the air to be cooled from below. This prevents the ground from warming the adjacent air layer, thereby suppressing the thermal convection and vertical mixing that would typically characterize an unstable or neutral profile. Furthermore, the fact that the wind came from the east over the snow-covered land rather than the relatively warmer waters of the Roskilde Fjord, ensured that the air mass remained cold and dense near the surface. The lack of surface heating explains why the measured lapse rate of -7.22 K/km was less negative than the dry adiabatic rate, as the cold surface environment stop the temperature from decreasing as rapidly with height as it would in a neutral or unstable condition.

In addition to that, the observed high turbulence intensity (TI ≈ 0.22) is consistent with this eastern wind direction. The terrain to the East has a higher roughness due to buildings and vegetation. This land-based flow can lead to mechanical turbulence through obstacles and friction, which explains the high TI values despite the thermally stable conditions that would otherwise suppress vertical mixing.

2 Part 2: Statistical analysis of multi-year wind climate

2.1 Data Description

Task: Describe the data. What has been measured, in what units, at which locations, with which instruments and with what temporal resolution?

The dataset consists of seven years of wind observations from 2018 to the end of 2025, provided as 10-minute mean values. The data in the file `tenMinutesData_2018_2024.csv` originates from the meteorological mast at the DTU Risø Campus.

Parameter	**Symbol**	**Unit**	**Instruments / Label**
Wind speed	sp	[m/s]	Cup (W) and Sonic (SW) anemometers
Wind direction	dir	[°]	Wind vanes (W) and Sonic (SW) sensors
Precipitation	Rain	[mm]	Tipping bucket rain gauge

Table 2: Summary of measured parameters and sensor types.

Measurements are available from different heights. From the 57m and 70m hight, SWsp, Wsp and Sdir are available. In addition to that the wind direction for 41m (Wdir) and data for the Rain are in the included in the data file.

2.2 Raw Data and Gaps

Task: Plot the raw data as function of time. Identify any data gaps and discuss possible reasons for these gaps.

2.2.1 Time Series Visualization

The raw 10-minute average data for wind speed, direction, and precipitation is plotted for the period from 2018 to the end of 2025. This includes measurements from cup anemometers, sonic anemometers, and wind vanes at various heights.

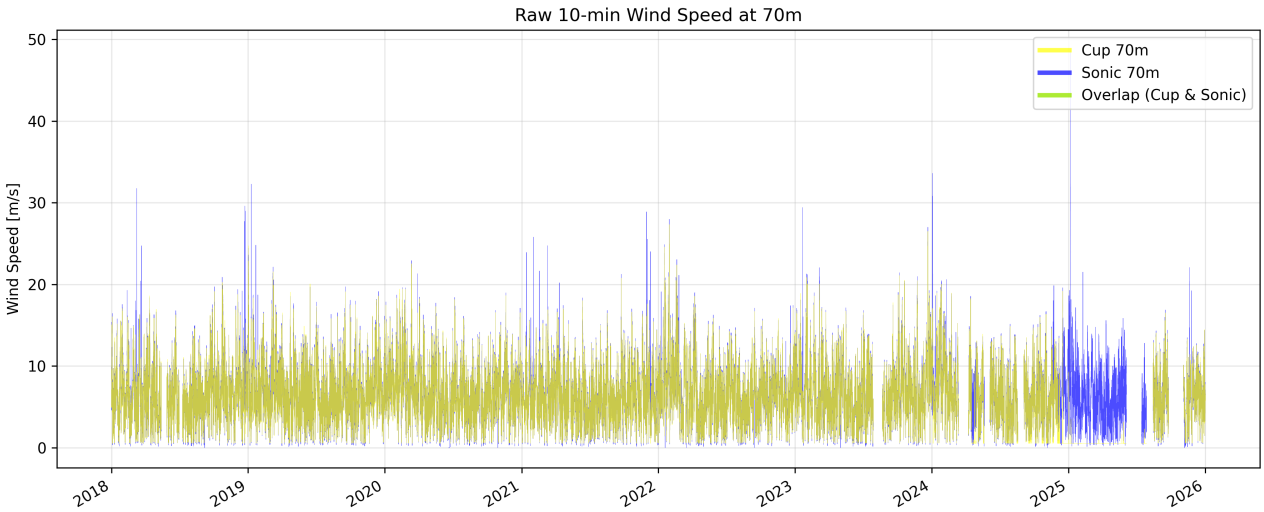

Figure 6: Raw 10-minute wind speed at 70m from the Cup and Sonic anemometers.

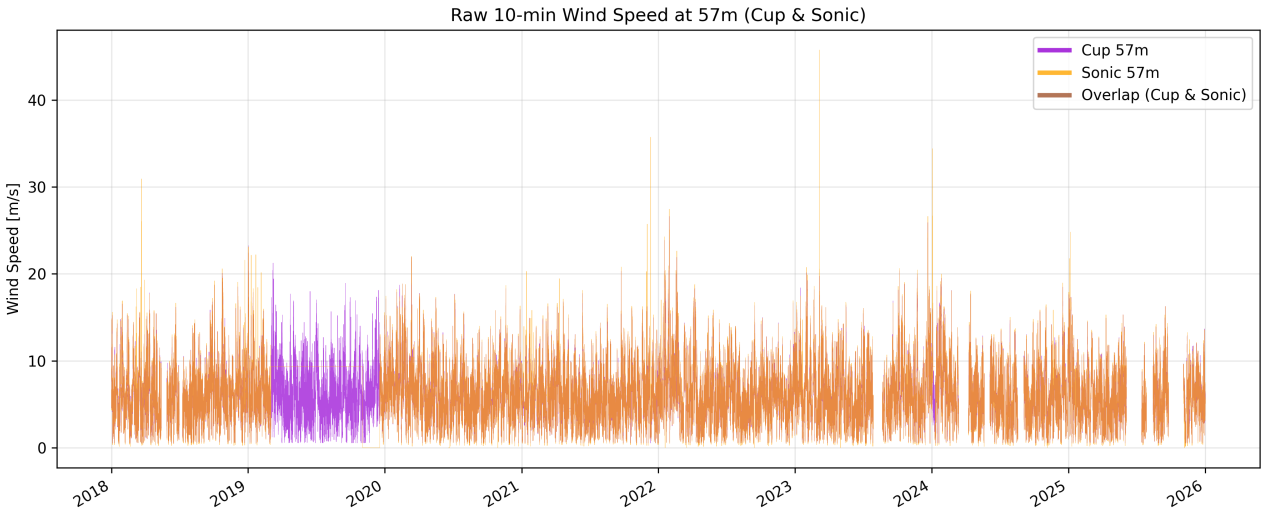

Figure 7: Raw 10-minute wind speed at 57m from the Cup and Sonic anemometers.

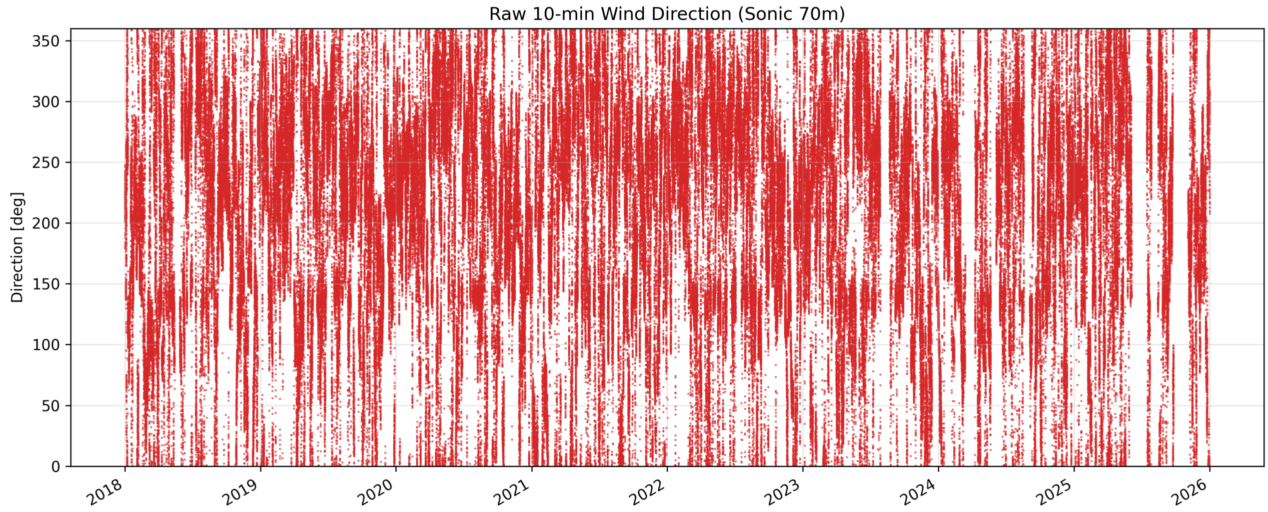

Figure 8: Wind direction time series from the Sonic anemometer at 70m height.

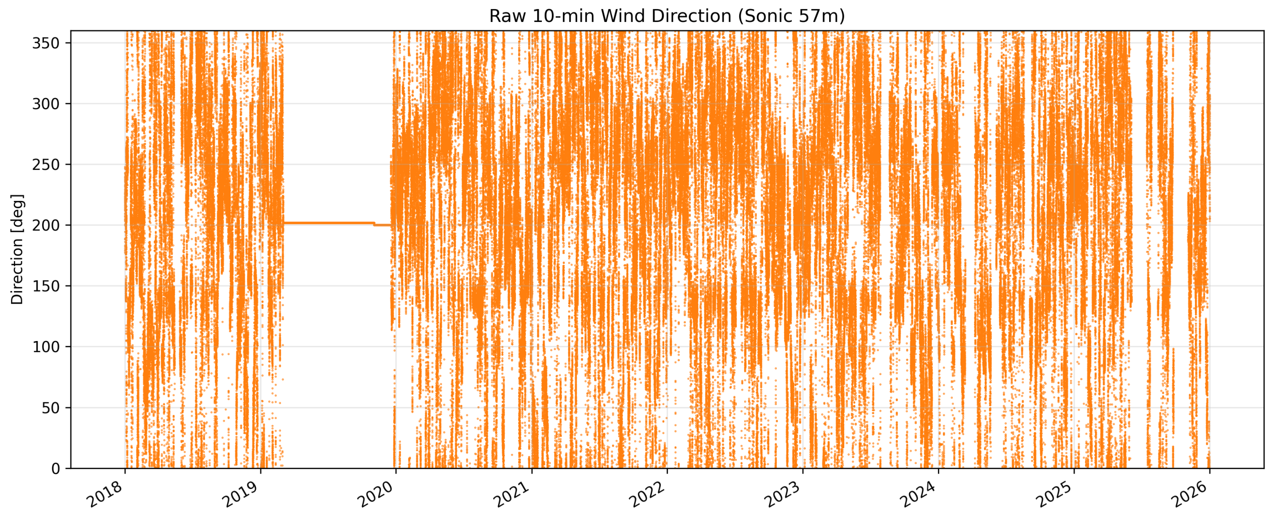

Figure 9: Wind direction time series from the Sonic anemometer at 57m height.

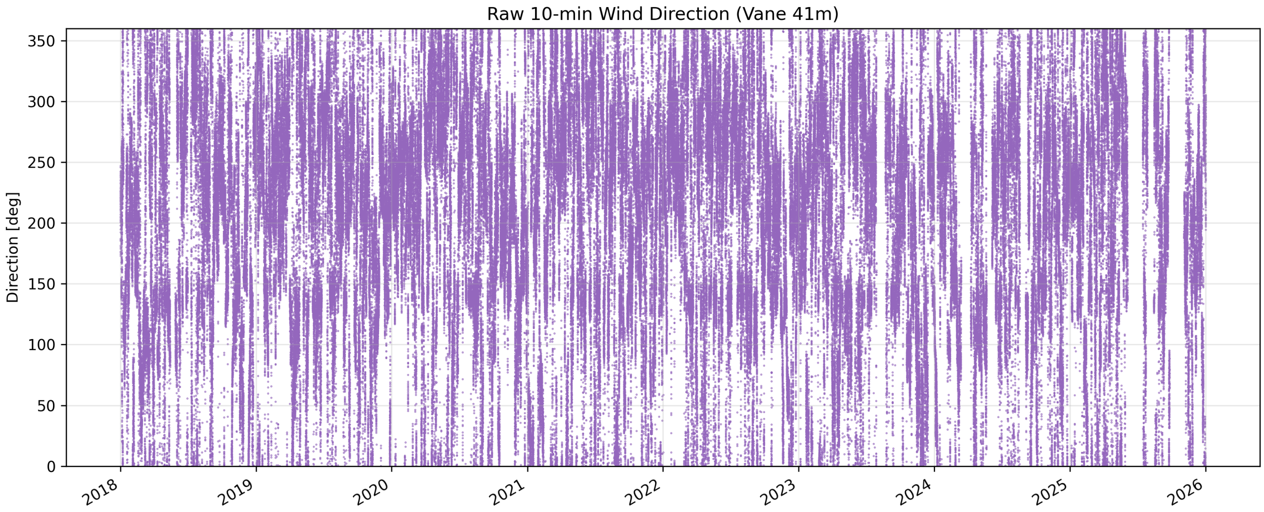

Figure 10: Wind direction time series from the wind vane at 41m height.

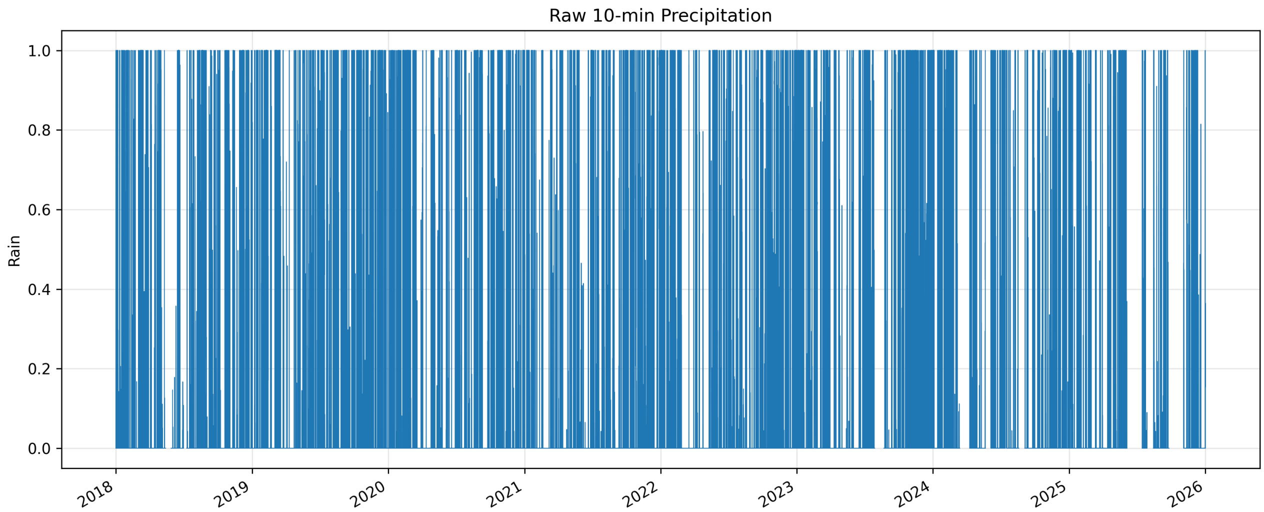

Figure 11: Time series of 10-minute precipitation (Rain) levels across the measurement period.

2.2.2 Data Gap Analysis and Discussion

To further investigate and assess the data quality, time intervals over the expected 10-minute resolution are identified as gaps. The following figures show the temporal distribution and duration of the missing records.

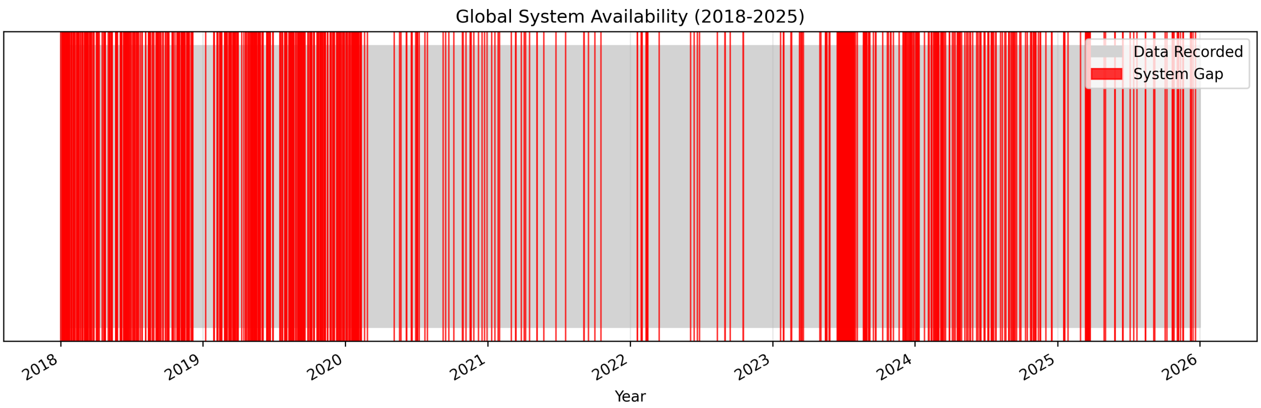

Figure 12: Global data availability from 2018 to 2026. Red segments identify total system outages where no timestamps were recorded.

The data gaps are predominantly in the years between 2018 and 2020. From earlier plots one can see that there is missing data of the wind speed at 70m hight in the first half of 2025 from the cup sensor. In addition to that, the cup snesor at 57 m also has a data gap between 2019 and 2020.

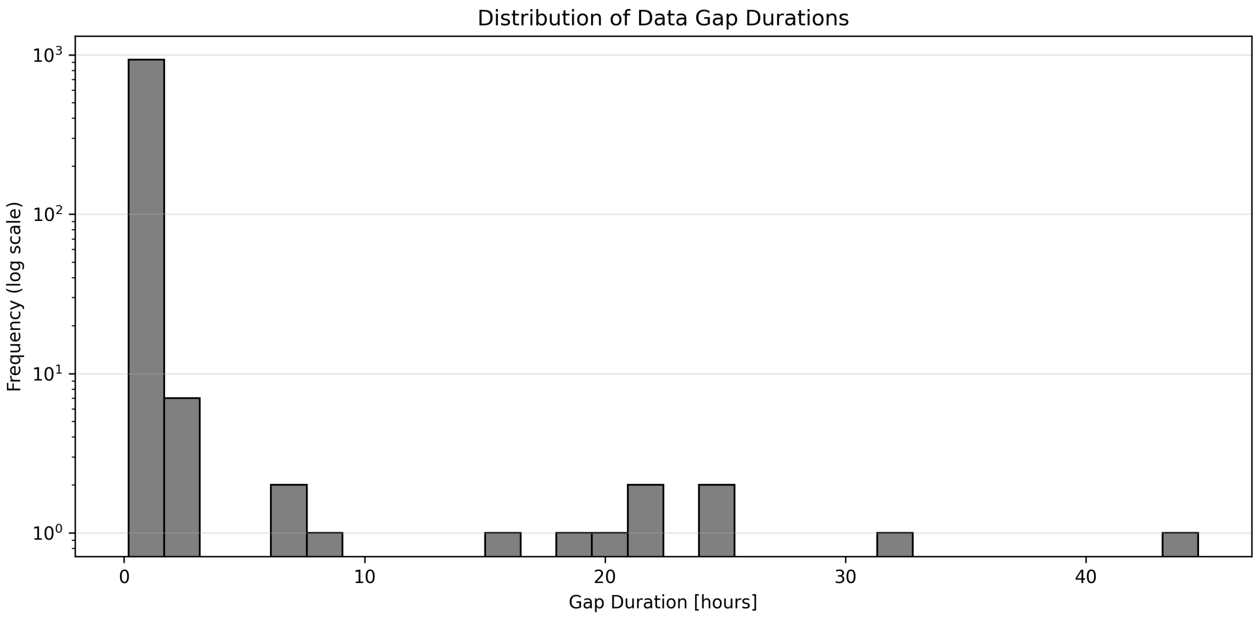

Figure 13: Distribution of gap durations on a logarithmic scale, showing the frequency of short versus long-term outages.

The short outtakes are dominating while longer outtages occur a lot less frequent. The longest outtage was around 45 hours.

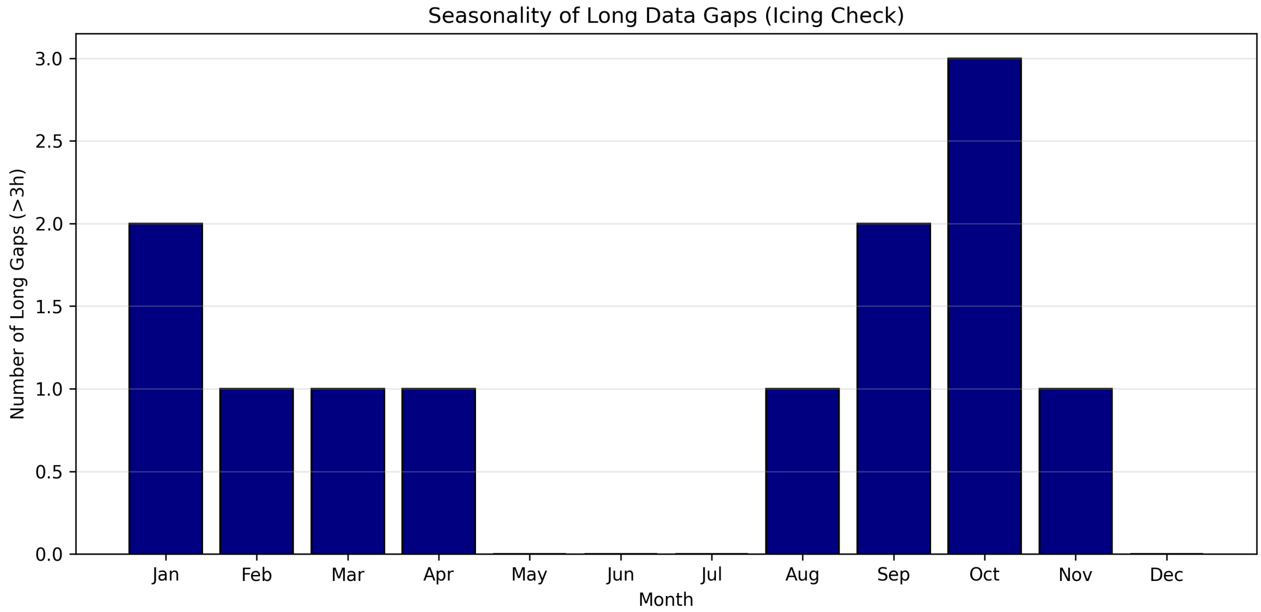

Figure 14: Seasonality of long-term gaps (> 3 hours), highlighting potential periods of sensor icing.

Discussion of Gaps: Possible reasons for the data gaps are:

- **Sensor Icing:** Seasonal analysis shows long-term gaps (> 3 hours) mainly from January to April. This indicates icing of cup anemometers and ultrasonic paths, which occurs even while the data logger remains powered.
- **Telemetry Errors:** The high frequency of short-duration gaps (10–20 minutes) in the histogram suggests transient signal loss during wireless data transmission from the mast to the server.
- **Local Hardware Defects:** Discrepancies between sensors at different heights (e.g., the 57m direction outage in 2019-2020) point to localized cable damage or specific instrument failure rather than a global system crash.
- **Scheduled Maintenance:** Simultaneous gaps across all sensor channels during summer months are typical for planned annual inspections, sensor calibrations, or manual data retrieval.
- **Atmospheric Interference:** Increased short-term gaps during summer afternoons correlate with lightning activity and electromagnetic interference from storms, potentially triggering automated system reboots.
- **Biological Factors:** Transient blockages of sonic anemometer paths or mechanical interference with cup anemometers by birds can lead to rejected 10-minute averages.
- **Power Supply:** System-wide outages (red bars in the availability plot) may result from battery voltage drops in the solar-powered system during periods of low solar radiation.

2.3 Seasonal Wind Roses

Task: Using the whole time series, plot a wind rose for the winter months (Dec-Feb) and one for the summer months (Jun-Aug) discuss the directional distribution of the wind. Does it match what you would expect for Denmark?

2.4 Histogram and Weibull Fit

Task: Define wind bins with a spacing of 1m/s and plot a histogram of the wind speed variation based on the data from 70m height for two different years, 2018 and 2024. Fit a Weibull distribution to the observed wind speed.

$$f(u) = \frac{k}{A}\left(\frac{u}{A}\right)^{k-1} \exp\left[-\left(\frac{u}{A}\right)^{k}\right] \tag{1}$$

2.5 AEP for V52 Turbine

Task: Calculate the AEP for the V52 turbine located at the met mast site, based on the 2018 and 2024 wind measured data. Discuss implications on the yearly variability.

Year	AEP [MWh]	Capacity Factor [%]
2018	...	...
2024	...	...

Table 3: Annual Energy Production (AEP) and Capacity Factor for the V52 turbine.